Working Memory Capacity and Everyday Studying

When people say they can only keep a few things in their head at once, they are describing something psychologists have measured for decades. Miller (1956) famously suggested that we hold about seven items, give or take two, although he was more careful about that number than the popular version suggests. Later work by Cowan (2001) argued that the real limit is closer to four chunks once rehearsal and grouping are controlled.

Baddeley and Hitch (1974) changed the question from how much to how: their model splits working memory into a phonological loop, a visuospatial sketchpad and a central executive. For studying, this matters because reading notes aloud and looking at a diagram load different parts of the system. In my own revision I noticed that I remember lecture slides better when I redraw the figures than when I reread the bullet points, which fits the idea of using two channels instead of one.

References

Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. Psychological Review, 63(2), 81-97. doi:10.1037/h0043158

Baddeley, A. D. and Hitch, G. (1974). Working memory. Psychology of Learning and Motivation, 8, 47-89. https://doi.org/10.1016/S0079-7421(08)60452-1

Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. Behavioral and Brain Sciences, 24(1), 87-114. https://doi.org/10.1017/S0140525X01003922